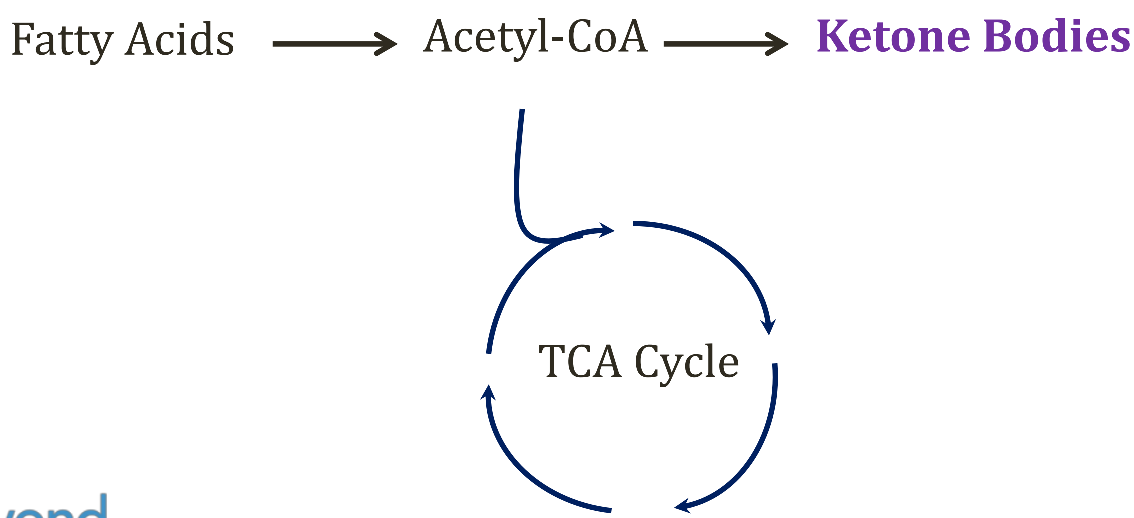
# Ketone Bodies

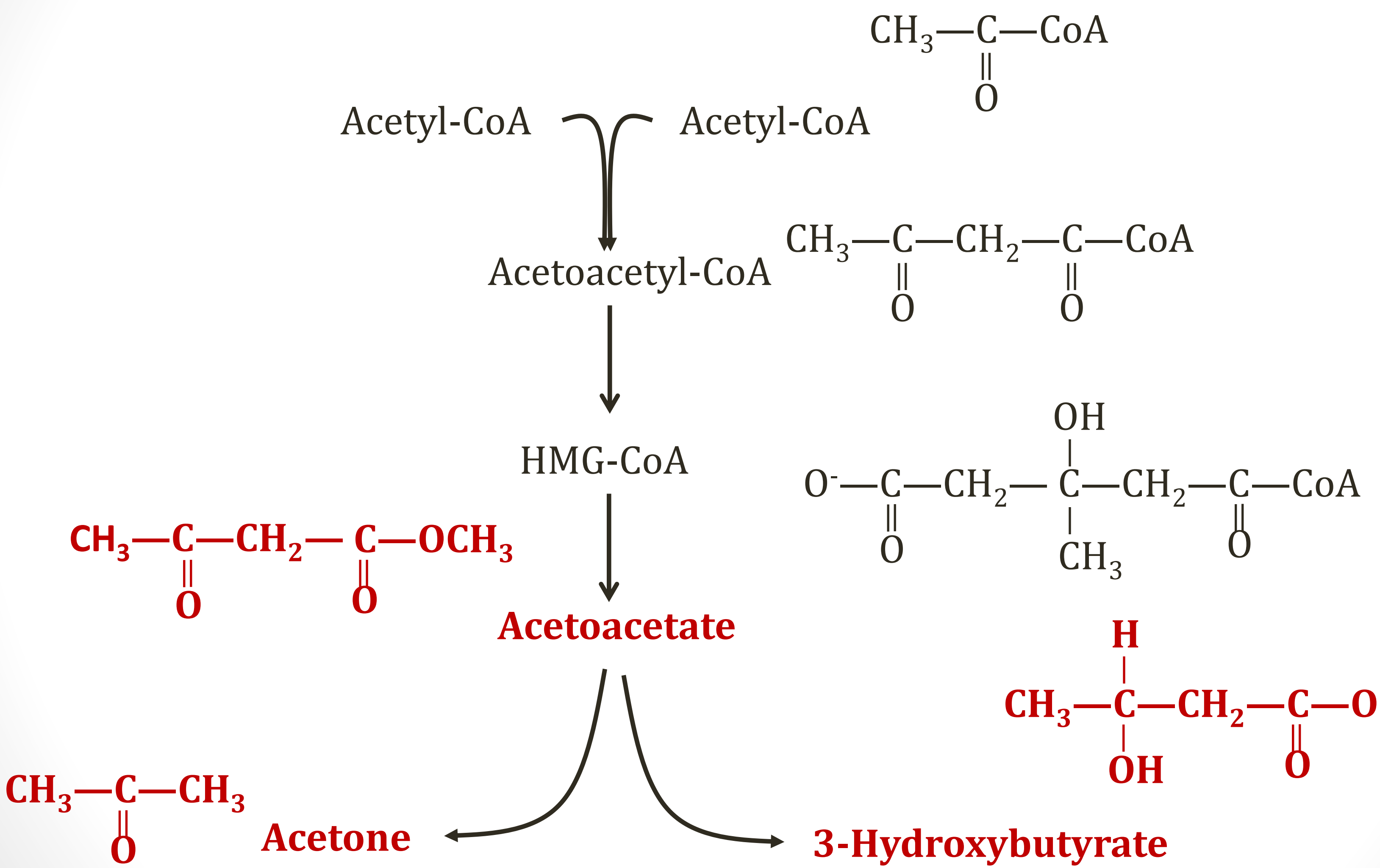
Jason Ryan, MD, MPH

# Ketone Bodies

- Alternative fuel source for some cells
- **Fasting/starvation** → fatty acids to **liver**
- Fatty acids → acetyl-CoA
- ↑ acetyl-CoA exceeds capacity TCA cycle
- Acetyl-CoA shunted toward ketone bodies



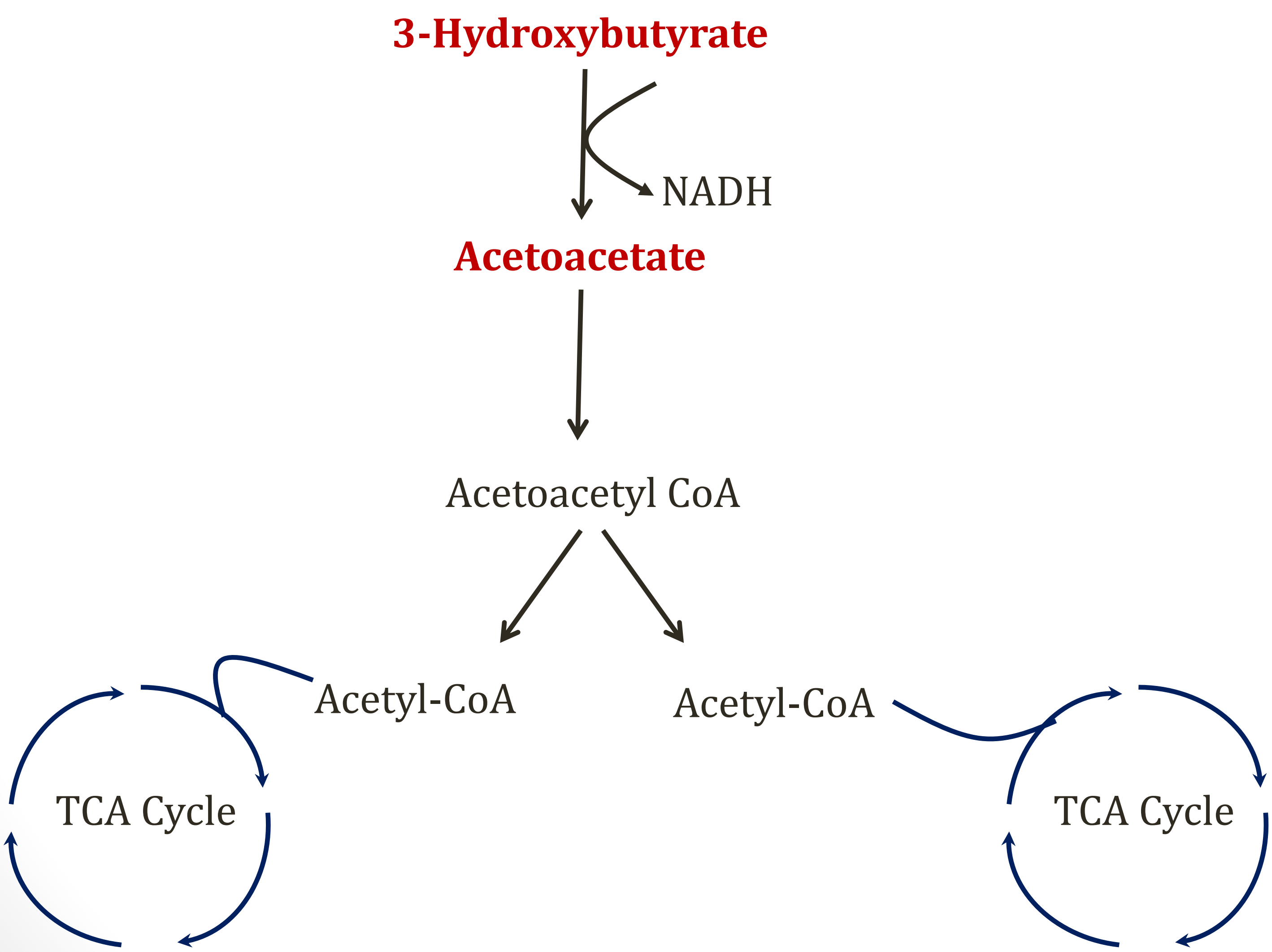
# Ketone Body Synthesis



# Ketolysis

- 3-hydroxybutyrate/acetoacetate → ATP
- Liver releases ketones into plasma
  - Constant low level synthesis
  - ↑ synthesis in fasting when FA levels are high
- Used by **muscle, heart**
  - Spares glucose for the brain
- Brain can also use ketone bodies
- Liver **cannot** use ketone bodies

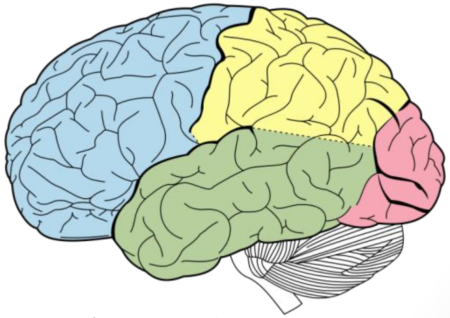
# Ketolysis


3-Hydroxybutyrate
NADH
Acetoacetate
Acetoacetyl CoA
Acetyl-CoA
Acetyl-CoA
TCA Cycle
TCA Cycle

# Big Picture

- Brain cannot use fatty acids
- Ketone bodies allow use of fatty acid energy by brain
- Also used by other tissues: preserve glucose for brain

Fatty Acids → Liver → Ketone Bodies → Brain → Acetyl-CoA

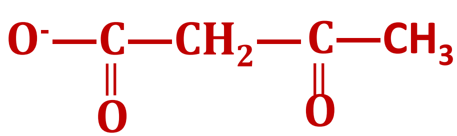


Schönfeld P, Reiser G. Why does brain metabolism not favor burning of fatty acids to provide energy? *Journal of Cerebral Blood Flow & Metabolism* (2013) **33,** 1493–1499

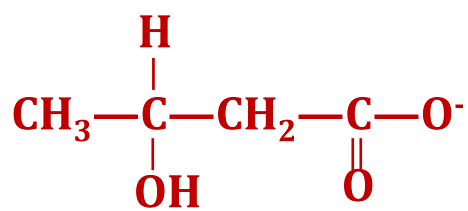
# Ketoacidosis

- Ketone bodies have low pKa
- Release $H^+$ at plasma pH
- ↑ ketones → anion gap metabolic acidosis

**Acetoacetate**

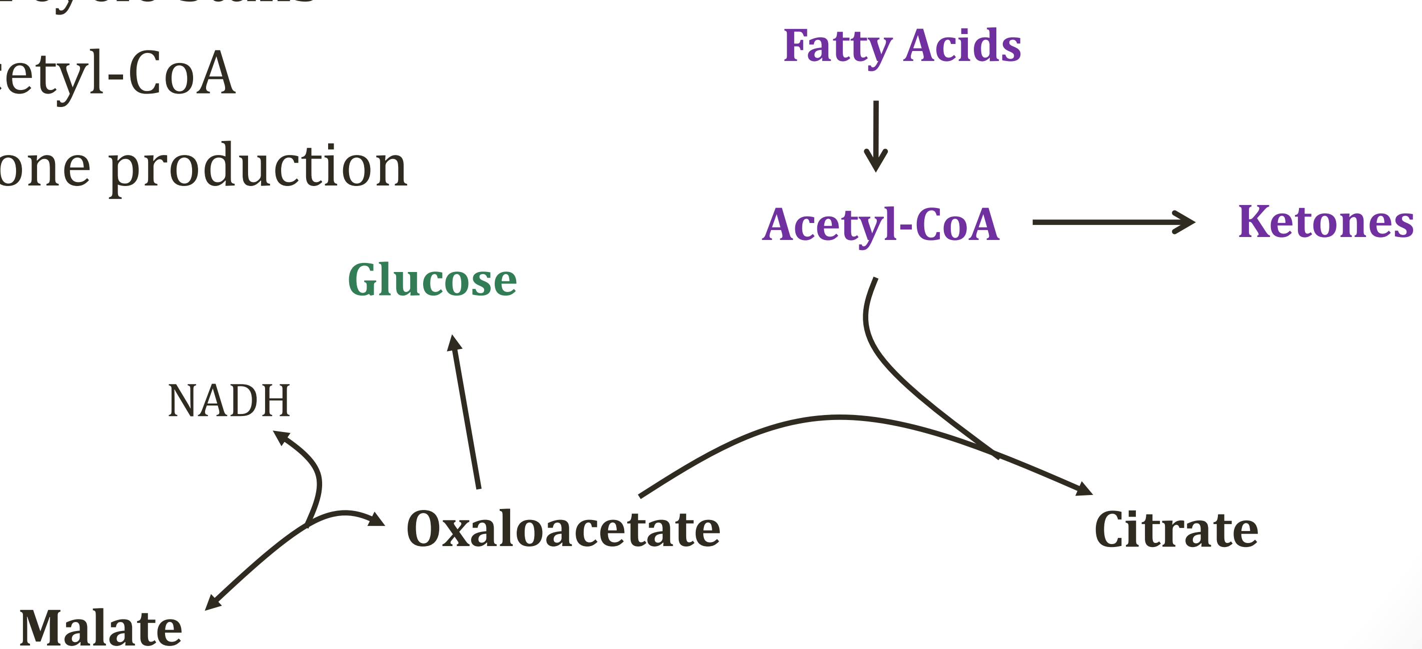



**3-Hydroxybutyrate**



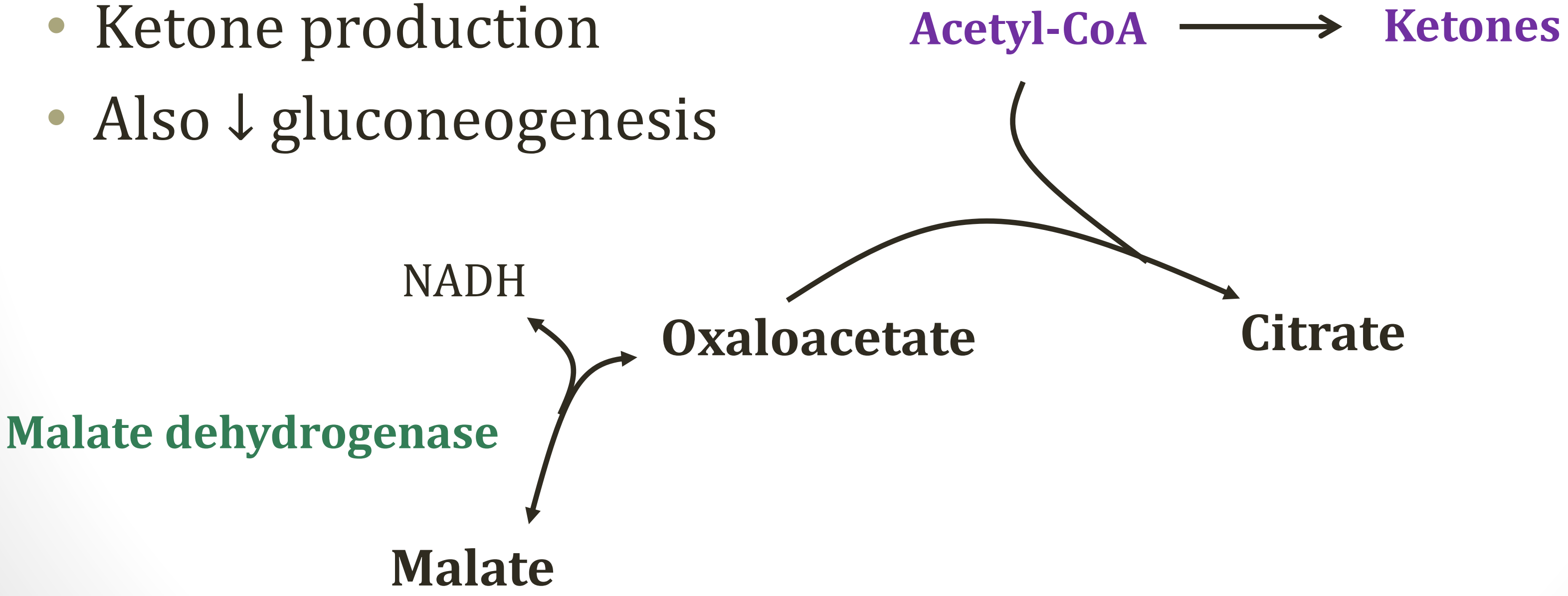
# Diabetes

- Low insulin
- High fatty acid utilization
- Oxaloacetate depleted
- TCA cycle stalls
- ↑ acetyl-CoA
- Ketone production



# Alcoholism

- EtOH metabolism: excess NADH
- Oxaloacetate shunted to malate
- Stalls TCA cycle
- ↑ acetyl-CoA
- Ketone production
- Also ↓ gluconeogenesis



# Urinary Ketones

- Normally no ketones in urine
  - Any produced →utilized
- Elevated urine ketones:
  - Poorly controlled diabetes (insufficient insulin)
  - DKA
  - Prolonged starvation
  - Alcoholism

Image courtesy of J3D3

# Ethanol Metabolism

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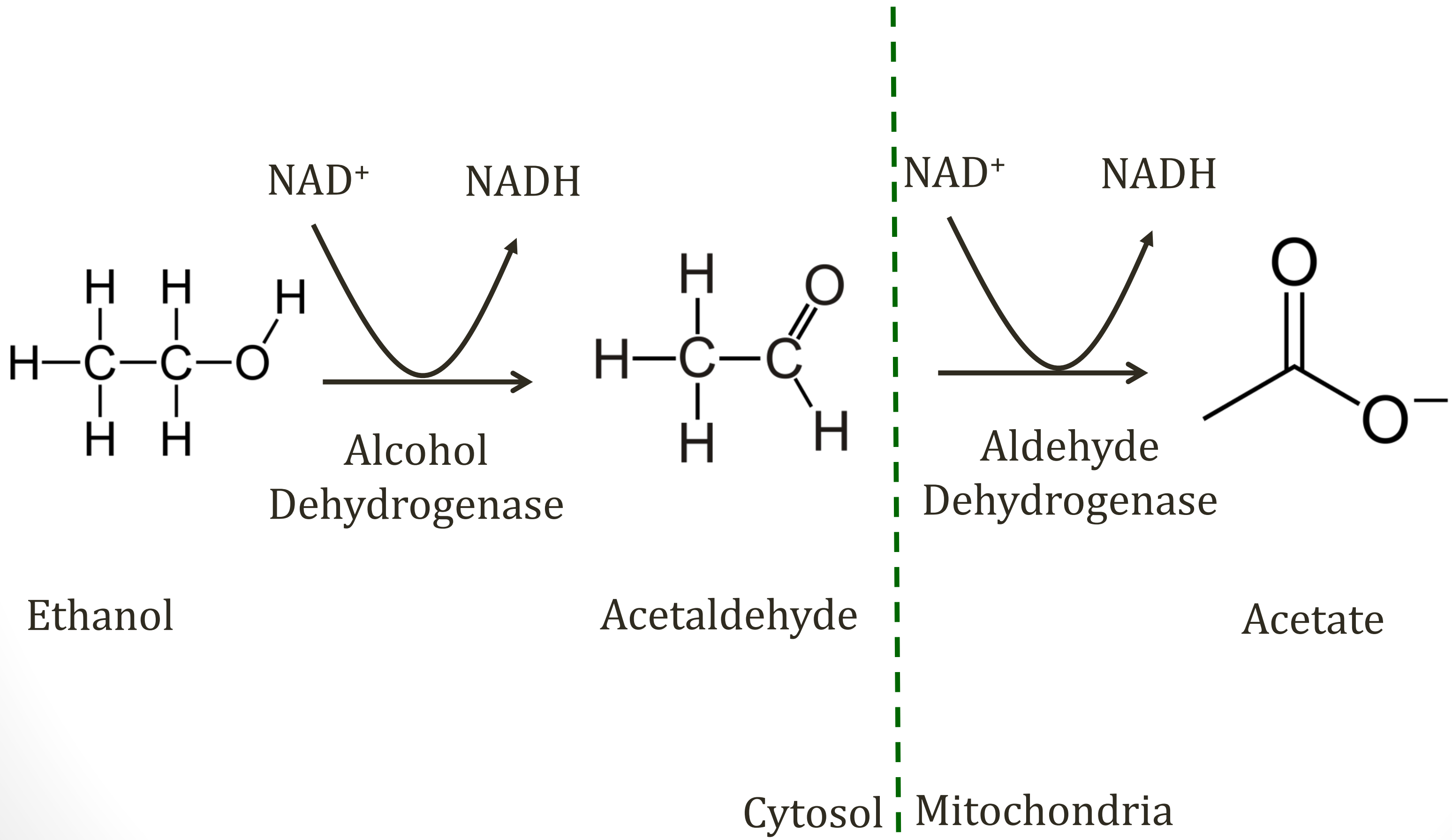
# Ethanol Metabolism

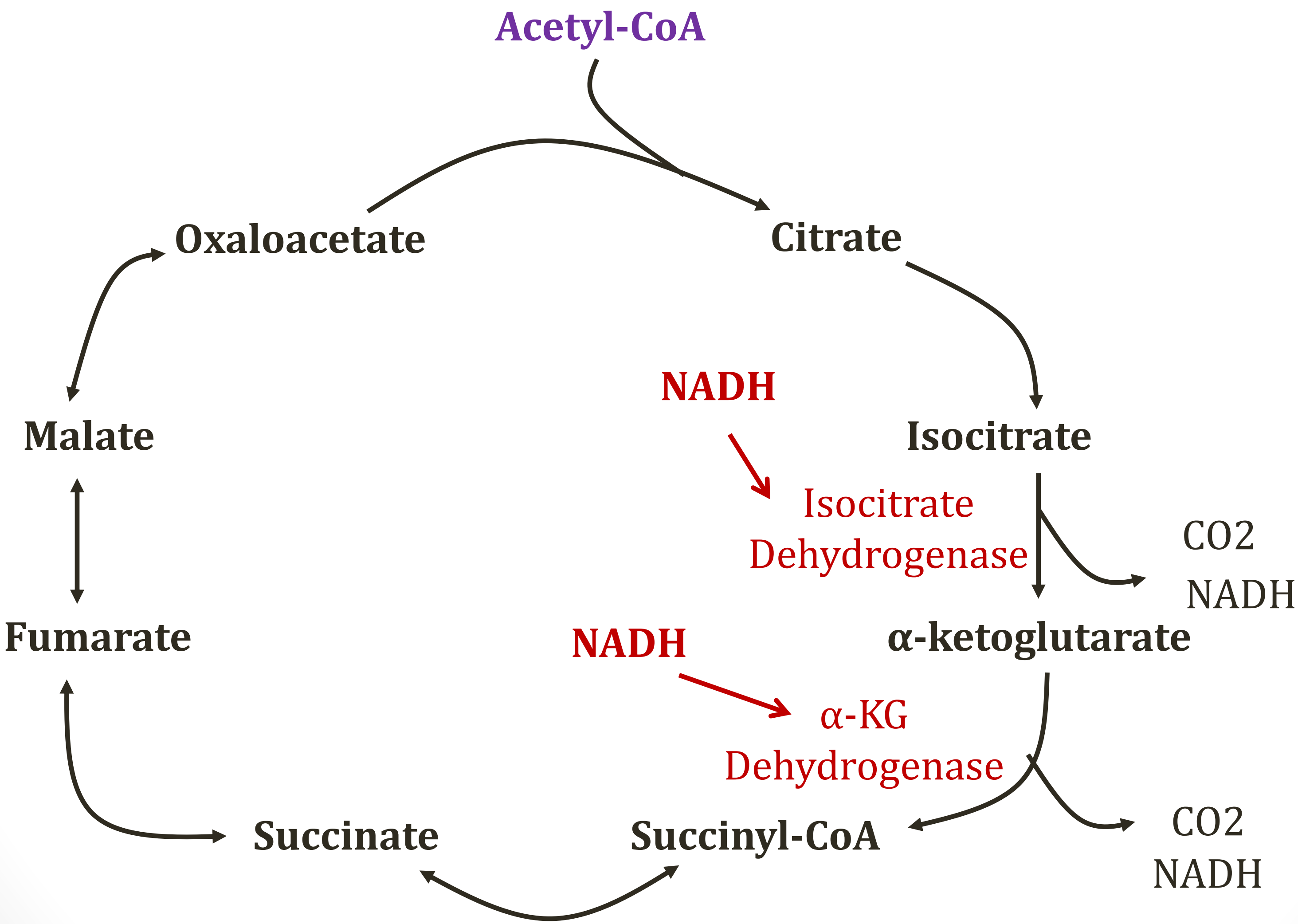


# Ethanol Metabolism

- Excessive alcohol consumption leads to problems:
  - CNS depressant
  - Hypoglycemia
  - Ketone body formation (ketosis)
  - Lactic acidosis
  - Accumulation of fatty acids
  - Hyperuricemia
  - Hepatitis and cirrhosis
- Trigger for all biochemical problems is **↑NADH**

Pixabay/Public Domain

# NADH Stalls TCA Cycle

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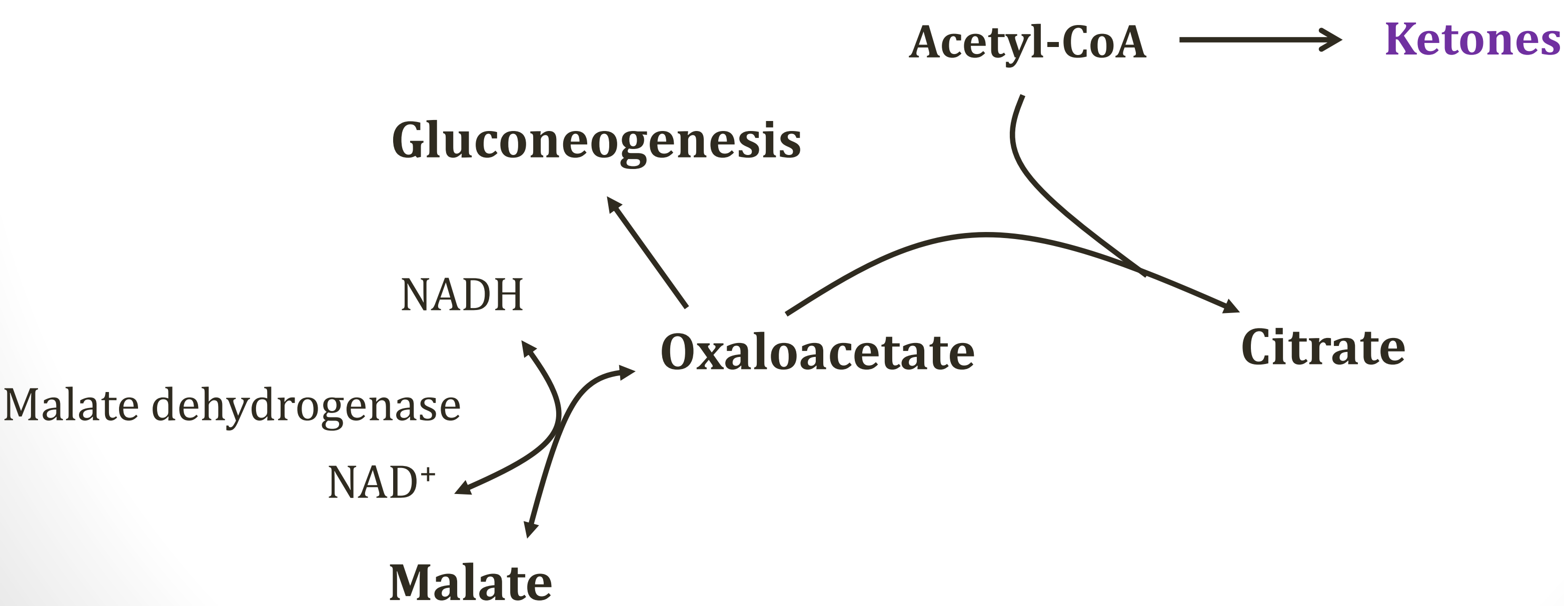
# NADH Stalls TCA Cycle

- NADH shunts oxaloacetate to malate
- ↑ acetyl-CoA
- **Ketone production**
- Also ↓ gluconeogenesis → **hypoglycemia**

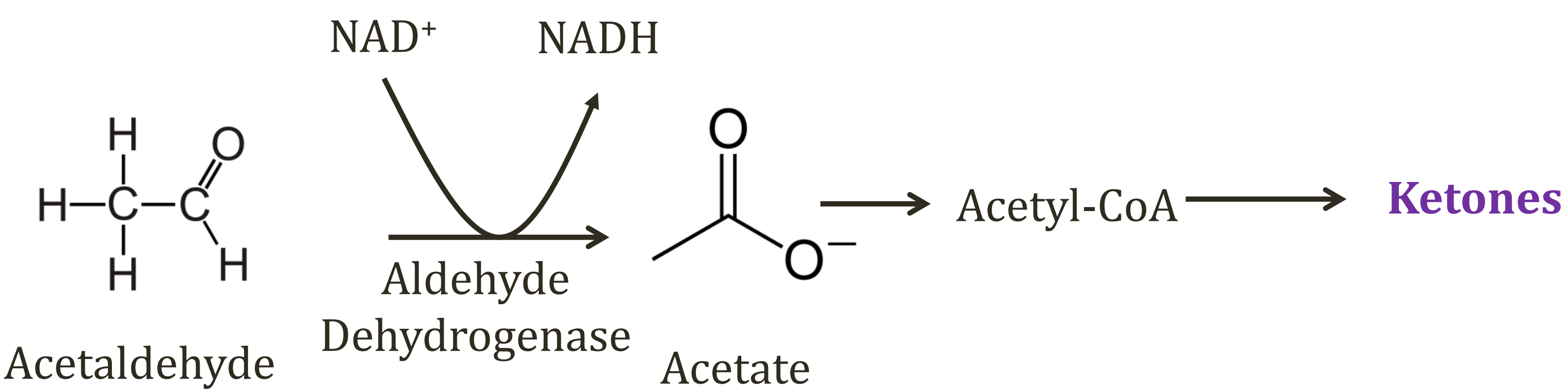


# Ethanol and Hypoglycemia

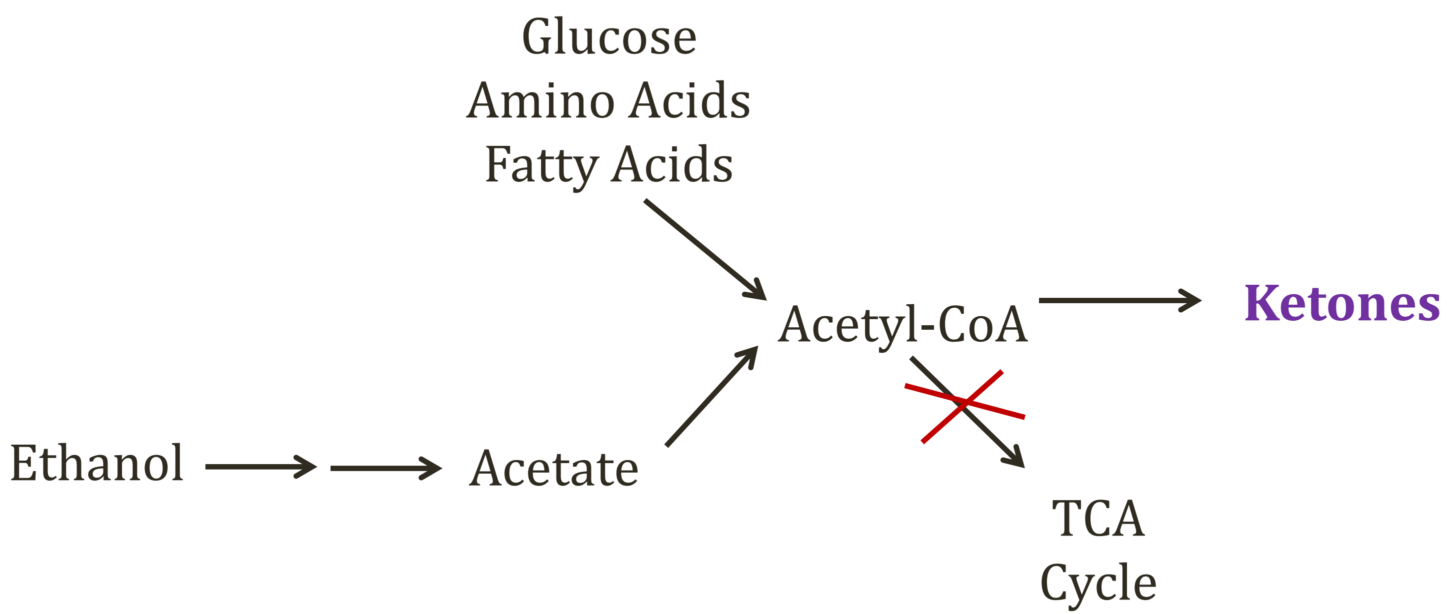
- Gluconeogenesis inhibited
  - Oxaloacetate shunted to malate
- **Glycogen** important source of fasting glucose
- Danger of low glucose when glycogen low
  - Drinking **without eating**
  - Drinking **after running**

# Ketones from Acetate

- Liver: Acetate → acetyl-CoA
- TCA cycle stalled due to high NADH
- Acetyl-CoA → ketones

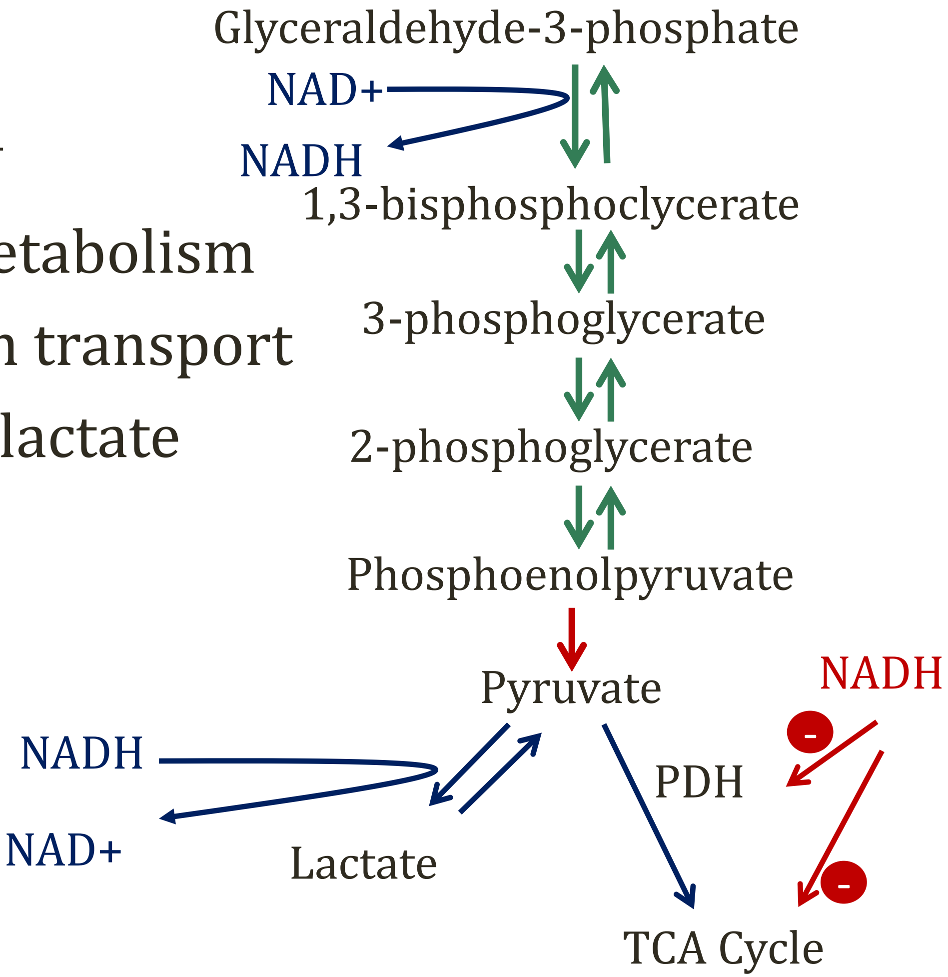


# Ketosis from Ethanol



# Lactic Acidosis

- Limited supply $NAD^+$
- Depleted by EtOH metabolism
- Overwhelms electron transport
- Pyruvate shunted to lactate
- Regenerates $NAD^+$



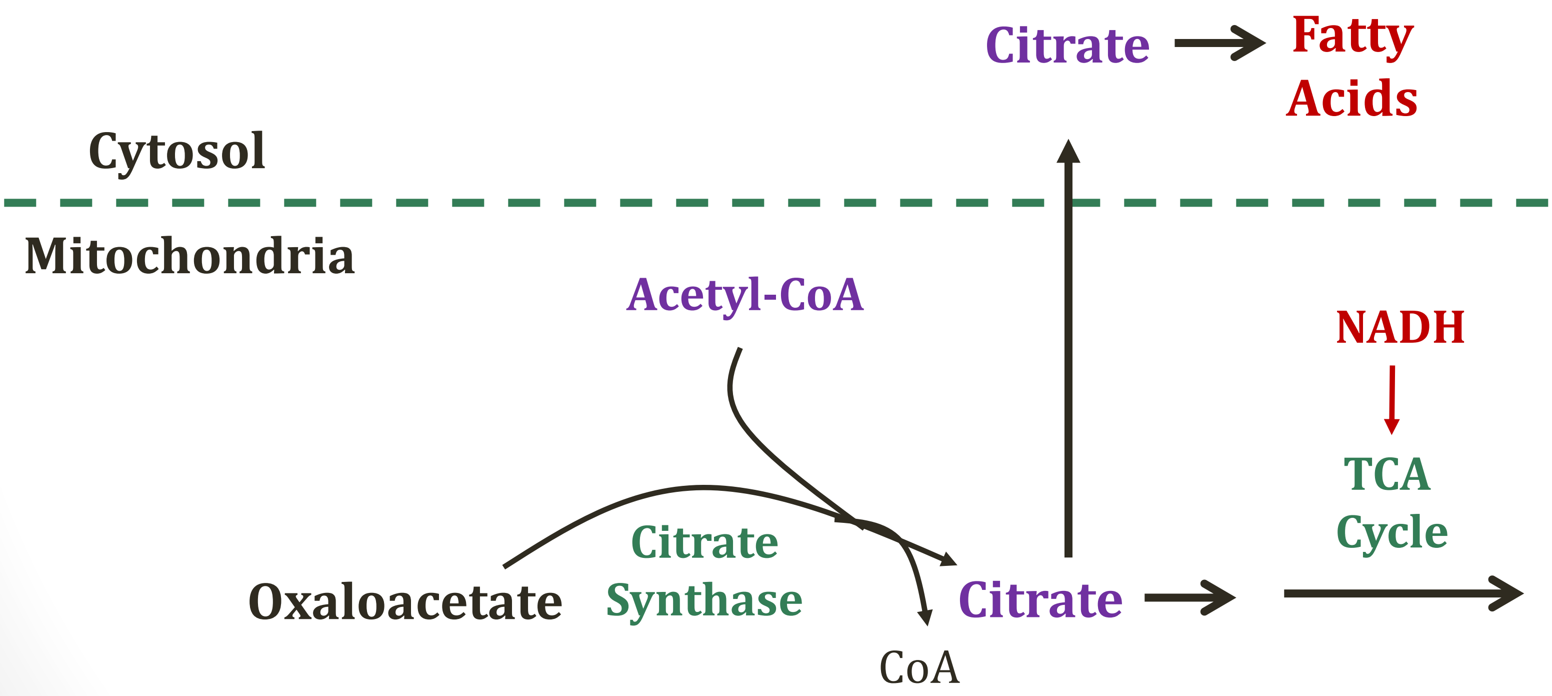
# Accumulation of Fatty Acids

- High levels NADH stalls beta oxidation
  - Beta oxidation generates NADH (like TCA cycle)
  - Requires NAD+
  - Inhibited when NADH is high
- Result: ↓ FA breakdown

β-oxidation

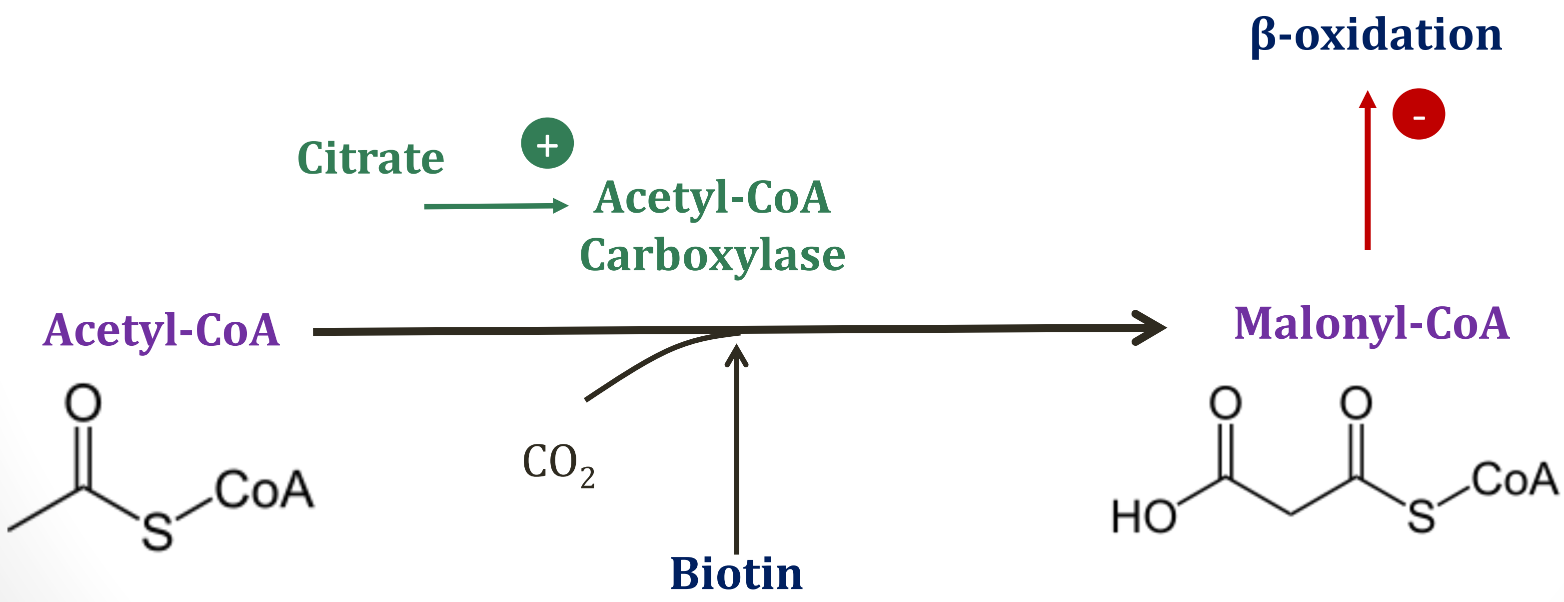
# Accumulation of Fatty Acids

- Stalled TCA cycle →↑ fatty acid synthesis



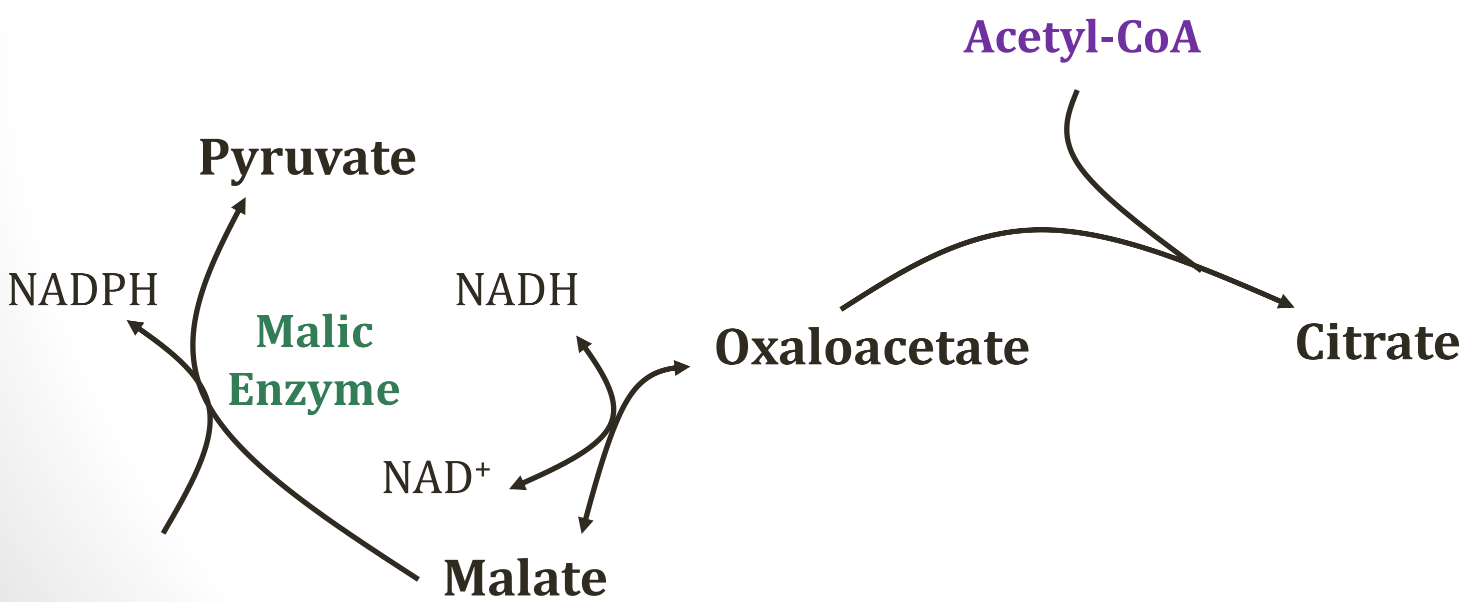
# Accumulation of Fatty Acids

- Rate limiting step of fatty acid synthesis
- Favored when citrate high from slow TCA cycle



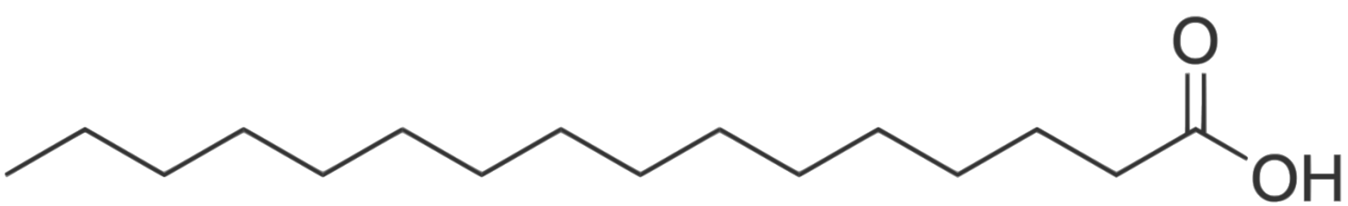
# Accumulation of Fatty Acids

- **Malate** accumulation also contributes to FA levels
- Used to generate **NADPH**
- NADPH favors fatty acid synthesis



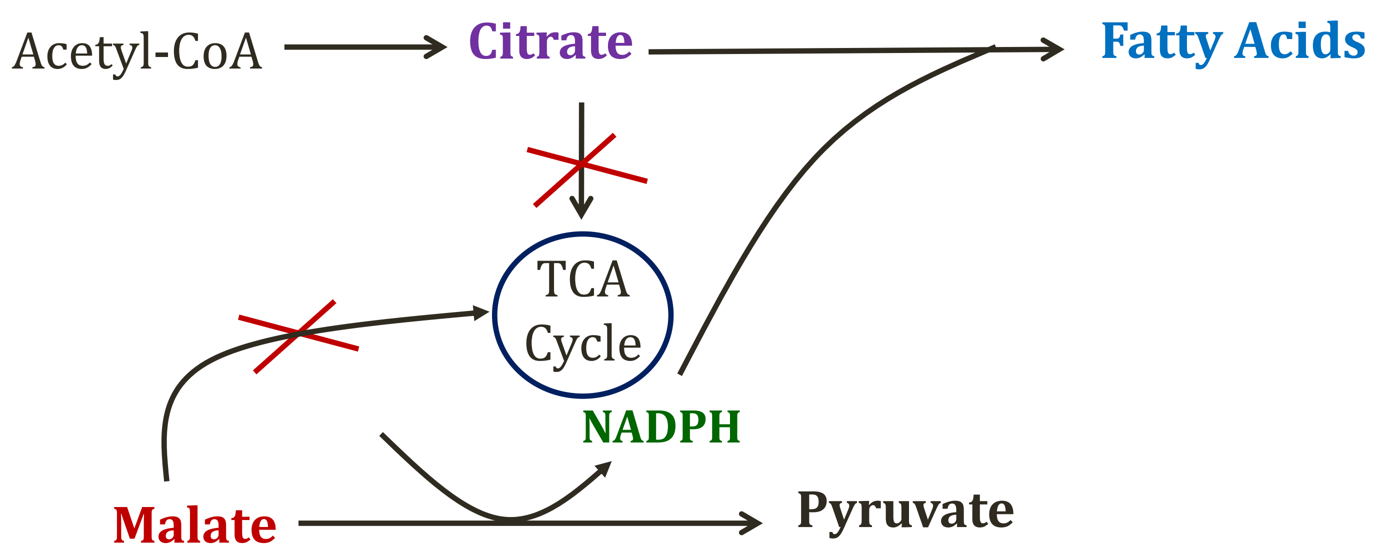
# Accumulation of Fatty Acids

- **Fatty acid synthase**
- Uses carbons from acetyl CoA and malonyl CoA
- Creates 16 carbon fatty acid palmitate
- Requires **NADPH**

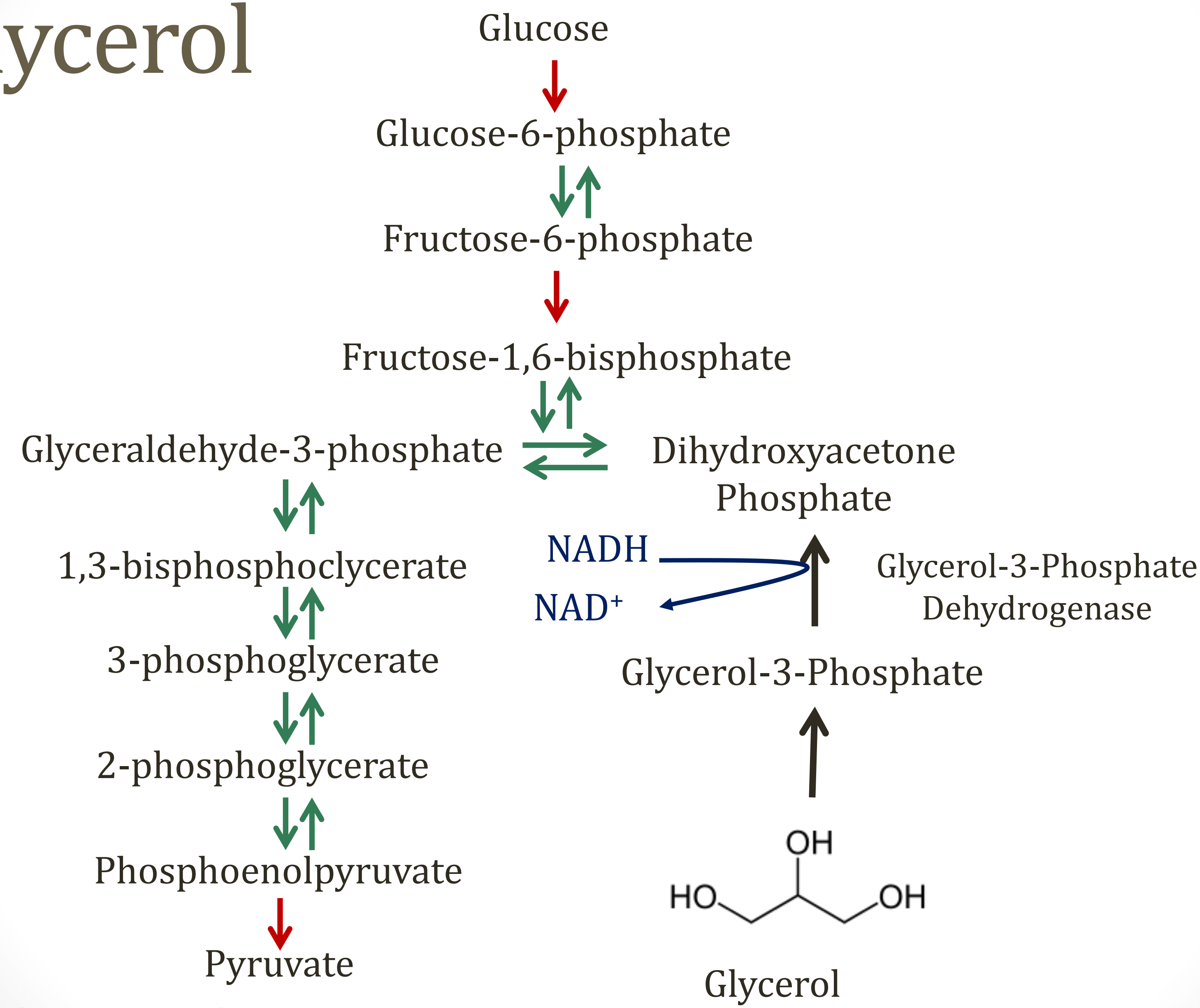



**Palmitate**

# Fatty Acids and Ethanol

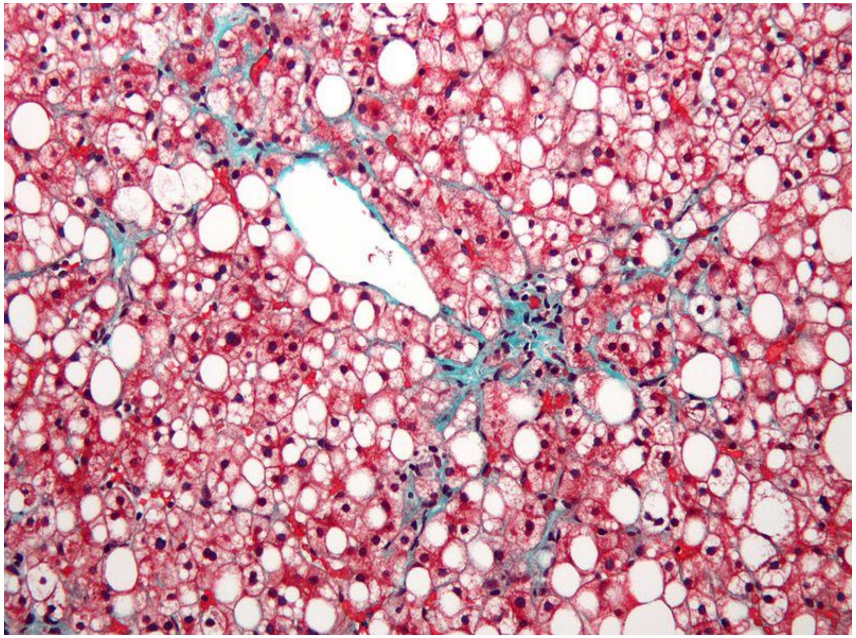


# Glycerol


Glucose
Glucose-6-phosphate
Fructose-6-phosphate
Fructose-1,6-bisphosphate
Glyceraldehyde-3-phosphate
Dihydroxyacetone Phosphate
1,3-bisphosphoclycerate
3-phosphoglycerate
2-phosphoglycerate
Phosphoenolpyruvate
Pyruvate
NADH
NAD+
Glycerol-3-Phosphate Dehydrogenase
Glycerol-3-Phosphate
OH
HO
OH
Glycerol

# Fatty Liver

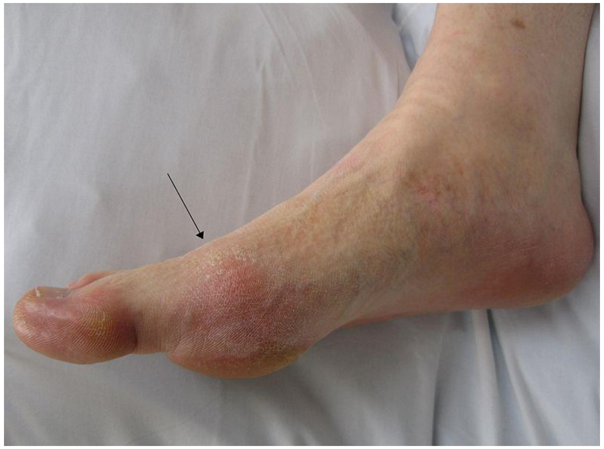
- Seen in alcoholism due to buildup of **triglycerides**



Nephron/Wikipedia

# Uric Acid

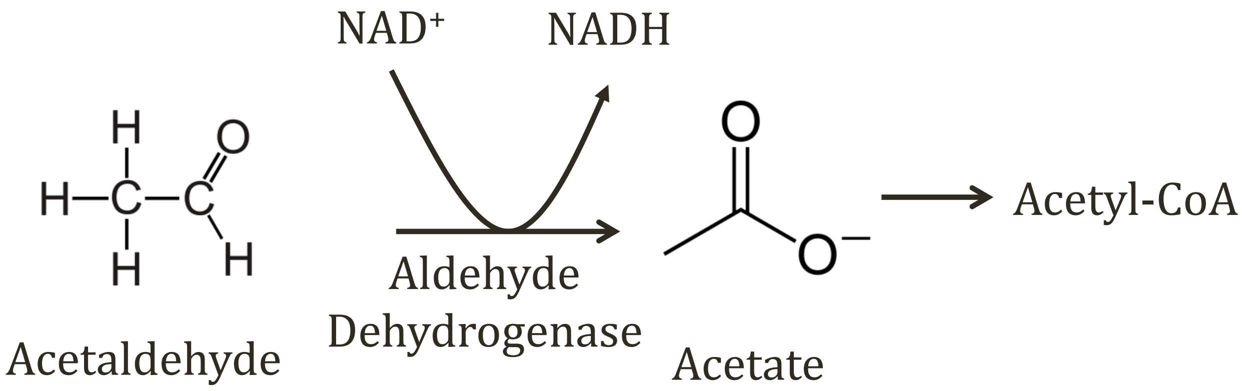
- Urate and lactate excreted by proximal tubule
- ↑ lactate in plasma = ↓ excretion uric acid
- ↑ uric acid → gout attack
- Alcohol a well-described trigger for **gout**



James Heilman, MD/Wikipedia

# Hepatitis and Cirrhosis

- High NADH slows ethanol metabolism
- Result: **buildup of acetaldehyde**
- Toxic to liver cells
- Acute: Inflammation → Alcoholic hepatitis
- Chronic: Scar tissue → Cirrhosis

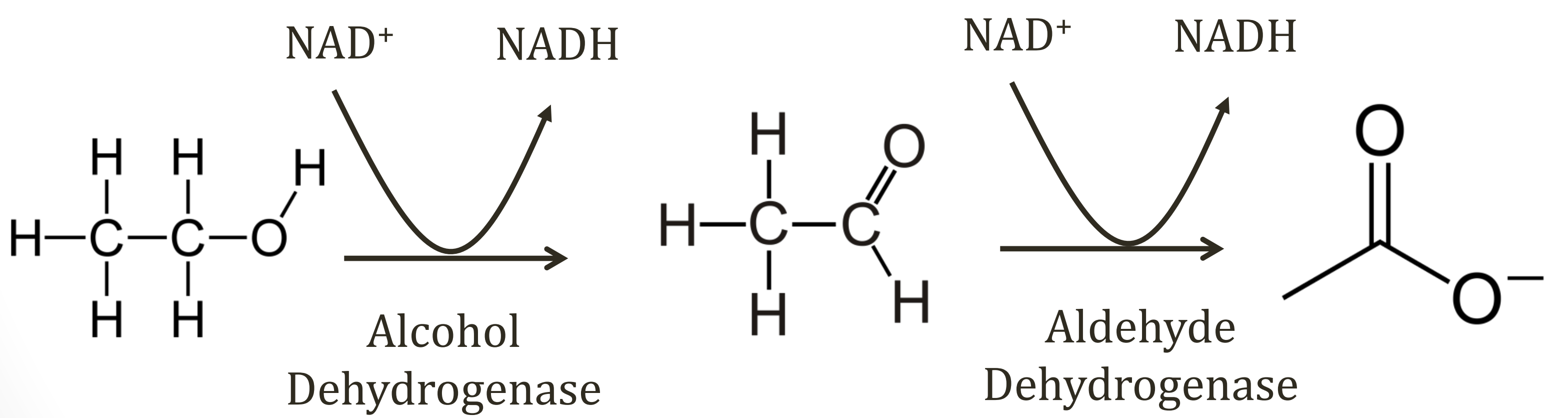


# Hepatitis and Cirrhosis

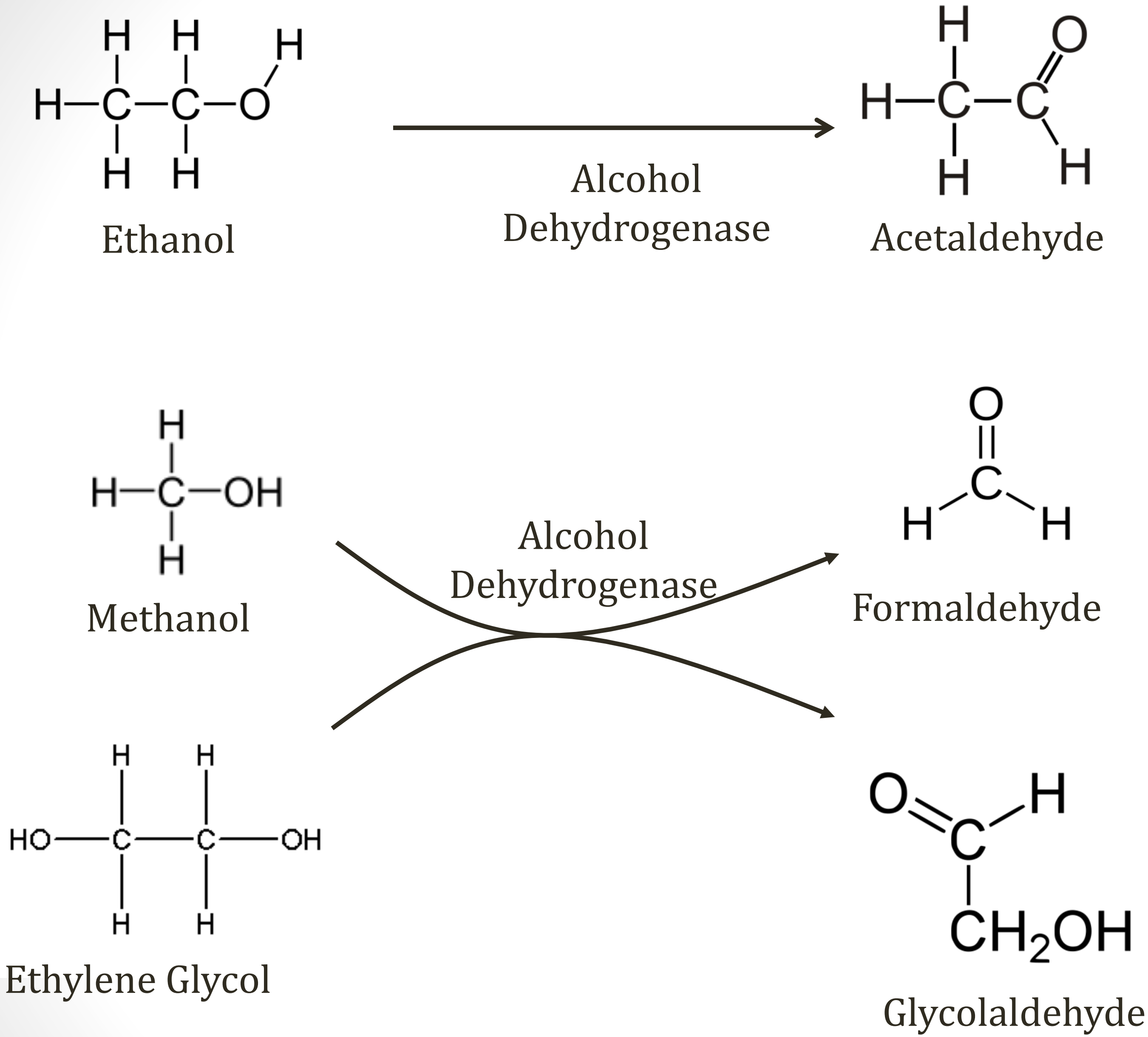
- **Microsomal ethanol-oxidizing system** (MEOS)
- Alternative pathway for ethanol
  - Normally metabolizes small amount of ethanol
  - Becomes important with excessive consumption
- Cytochrome P450-dependent pathway in **liver**
- Generates acetaldehyde and acetate
- Consumes NADPH and Oxygen
- Oxygen: generates free radicals
- NADPH: glutathione cannot be regenerated
  - Loss of protection from oxidative stress

# Alcohol Dehydrogenase

- Zero order kinetics (constant rate)
- Also metabolizes methanol and ethylene glycol
- Inhibited by fomepizole (antizol)
    - Treatment for **methanol/ethylene glycol** intoxication

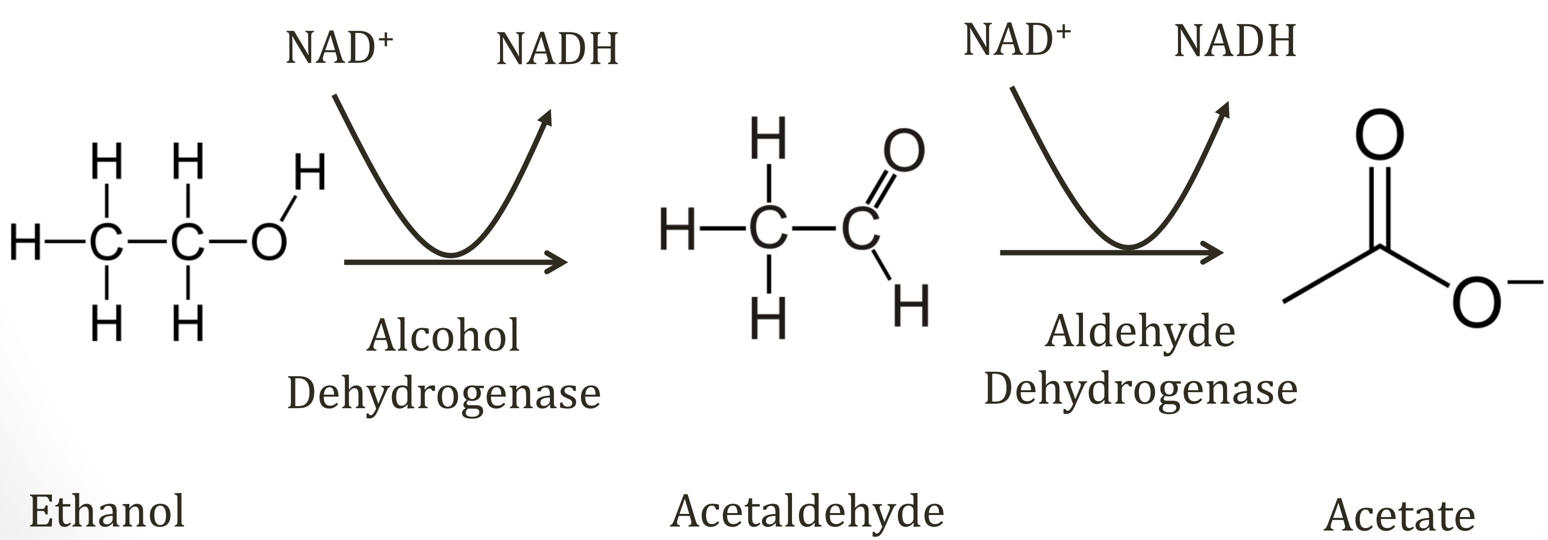



Ethanol Acetaldehyde Acetate


Ethanol
Alcohol Dehydrogenase
Acetaldehyde
Methanol
Alcohol Dehydrogenase
Formaldehyde
Ethylene Glycol
Glycolaldehyde

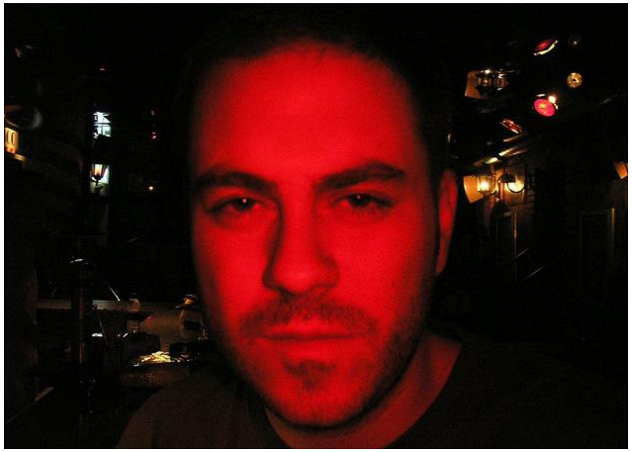
# Aldehyde Dehydrogenase

- Inhibited by disulfiram (antabuse)
- Acetaldehyde accumulates
- Triggers catecholamine release
- **Sweating, flushing**, palpitations, nausea, vomiting



# Alcohol Flushing

- Skin flushing when consuming alcohol
- Due to slow metabolism of acetaldehyde
- Common among Asian populations
  - Japan, China, Korea
  - Inherited deficiency aldehyde dehydrogenase 2 (ALDH2)
- Possible ↑risk esophageal and oropharyngeal cancer



**Jorge González**/Flikr